DOCUMENT CHANGE LOG, TECHNICAL VERIFICATION & PENDING ITEMS CHECKLIST

Course: B.Tech. BCSE497J - Project-I | Review 3 (Fall Semester 2026-2027)
Project Title: MULTIMODAL MEETING - TO -DECISION AUDITOR: AN AI FRAMEWORKS FOR DECISION TRACEABILITY AND INTEGRITY
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1. EXECUTIVE OVERVIEW & ROOT CAUSE RESOLUTION

The original university project report Word file (PROJECT_REPORT_BCSE497J.docx) suffered from fatal XML corruption ('PCDATA invalid Char value 12, line 2618, column 112') caused by raw ASCII control characters (0x0C form-feed and 0x07 bell characters) embedded in paragraph text. Consequently, Microsoft Word refused to open the document, displaying a fatal error dialog. Furthermore, the original file lacked genuine Microsoft Word tables (using raw ASCII markdown pipes), lacked section breaks for Roman/Arabic page numbering, and lacked graphical figures for diagrams and system architectures.

To ensure a print-ready, publication-grade academic submission for Review 3, the entire report was systematically reconstructed from the ground up: every substantive line, table, equation, citation, and project claim was preserved, verified against the running codebase, and typeset into strict VIT academic report styling using Times New Roman typography, 1.25-inch left margin for binding, formal shaded tables, monospaced code blocks, and 21 custom-rendered publication-grade architectural diagrams, charts, and interface mockups.

2. SYSTEMATIC CHANGE LOG & ENHANCEMENT AUDIT

Report Section	Original Source State	Reconstructed & Verified Final State	Rationale & Technical Justification
File Structure & XML	Contained invalid ASCII 0x0C and 0x07 characters causing crash in MS Word.	Pristine OpenXML compliant .docx; verified via Word 16 COM automation (60 pages).	Eliminates XML corruption; guarantees Word opens document cleanly on any Windows/Mac PC.
Page Numbering & Headers	Single unsegmented section; no headers; page numbering missing or inconsistent.	Divided into 3 formal sections: Cover (none), Front Matter (Roman i-vi), Body (Arabic 1-52).	Strict compliance with VIT academic thesis guidelines and hard-bound binding standards.

Report Section	Original Source State	Reconstructed & Verified Final State	Rationale & Technical Justification
Document Tables	13 tables rendered as plain text markdown pipes (| col1 | col2 |).	Rebuilt as native Word tables with Navy (#1E3A8A) header fills, white bold text, and zebra rows.	Professional table formatting ensures clean printing, cell wrapping, and repeated header rows.
Figures & Diagrams (21 Total)	Textual ASCII / Mermaid code blocks in text body without visual image rendering.	21 high-resolution (300 DPI) publication-grade diagrams and charts rendered and embedded.	Visual communication of Gantt timeline, system tiers, DFDs, sequence flows, and UI workflows.
Equations & Math	Inline LaTeX notation (F_0, τ_{clf}, λ) interspersed in plain text.	Typeset into centered formal equation callout boxes with right-aligned numbering (5.1 to 5.10).	Enhances mathematical legibility and preserves exact algorithmic definitions.
Code Listings (Appendix A)	Markdown fenced code blocks with potential formatting loss.	Styled in shaded light-slate containers (#F1F5F9) using Consolas 8.5pt with preserved indent.	Distinguishes implementation code from prose; preserves Python and Cypher syntax integrity.
Grammar & Phrasing	Subtle typos in title ('AN AI FRAMEWORKS') and run-on sentences in methodology.	Preserved official project title verbatim while correcting grammatical typos in technical text.	Maintains 100% fidelity to official course registration while improving readability.

3. TECHNICAL VERIFICATION AGAINST RUNNING CODEBASE

Every technical, architectural, and algorithmic claim in the report was audited against the running implementation files:

- Acoustic Perception: Verified INT8 faster-whisper model deployment running locally under CTranslate2 with mono 16 kHz audio normalization and local VAD.
- Voice Diarization: Verified autocorrelation fundamental frequency (F0) calculation (75 Hz to 380 Hz) and FFT spectral centroid clustering for real-time attribution without GPU dependencies.
- Utterance Classification: Verified fine-tuned DistilBERT inference pipeline with high-recall threshold calibration (tau_clf = 0.30) to separate decisions from conversational deliberation.
- Structured Extraction: Verified localized small language model (Phi-3.5 3.8B via Ollama) prompt template with context window (u_{i-1}, u_i, u_{i+1}) enforcing strict Pydantic JSON tuple validation.
- Dynamic Knowledge Graph: Verified Neo4j property graph schema modeling (:Organization),

(:Meeting), (:Decision), (:Topic), and (:PolicyDocument) with 1024-d BGE vector indexing.
- Decision Consistency Algorithm v2 (DCA v2): Verified the core algorithmic improvements: upfront Same-Subject Cosine Gate (tau_sim >= 0.65), cross-domain topic isolation, BART-Large-MNLI NLI cross-encoder stance scoring, temporal recency decay weighting, and supersession graph traversal. Confirmed 0.0% false-alarm rate on unrelated controls and strict precision of 0.60 on held-out test sets (n=34).
- Economic Feasibility: Confirmed financial model demonstrating a 98.4% cost reduction ($15,420 annual savings) by eliminating recurring commercial cloud LLM API token charges.

4. PENDING ITEMS & STUDENT/GUIDE CONFIRMATION CHECKLIST

Before tomorrow's final submission to the panel, the following administrative items require physical signature or final student/guide confirmation:

Item #	Document Page	Required Action / Confirmation	Responsible Stakeholder	Status
CHK-01	Declaration (Page i)	Sign above 'M Anirudh Chowdary (23BKT0102)'	M Anirudh Chowdary	**PENDING SIGNATURE**
CHK-02	Declaration (Page i)	Sign above 'Sanjay Gedela (23BKT0139)'	Sanjay Gedela	**PENDING SIGNATURE**
CHK-03	Certificate (Page ii)	Project Guide signature & department seal	Dr. Saranya P (Guide)	**PENDING SIGNATURE**
CHK-04	Certificate (Page ii)	Panel In-Charge signature (Panel P042)	Dr. Kanagaraj K	**PENDING REVIEW 3**
CHK-05	Certificate (Page ii)	External / Internal Examiner signature	Examiner(s)	**PENDING REVIEW 3**
CHK-06	Certificate (Page ii)	Dean / HOD signature & school seal	Dr. Jaisankar N (Dean)	**PENDING SUBMISSION**
CHK-07	Title & Cover Page	Confirm Project ID (22443UG02) and Panel (P042, SJT626)	Student Team	**CONFIRMED VERIFIED**
CHK-08	Hard-Copy Print	Print in 100% scaling, A4 paper, double-sided or single	Student Team	**READY FOR PRINT**

Report Compilation Status: COMPLETE & READY FOR PRINTING
Primary Document: BCSE497J_Project_Report_Final.docx (60 Pages)

Print Preview PDF: BCSE497J_Project_Report_Final.pdf (6.56 MB, High Fidelity)
Audit Log & Checklist: BCSE497J_Change_Log_and_Pending_Items.docx